PROJECT PORTFOLIO / 2026

Voting System
// secure.elections

A Java + Oracle SQL desktop application for managing voters, candidates, elections and ballot integrity.

JAVA · SWING UI · JDBC · ORACLE SQL · DESKTOP

DOMAIN	STACK	ROLE	STATUS
Civic Tech / Elections	Java • Swing • JDBC • Ora	Full-stack Developer	v1.0 shipped

01 Project Overview

The **Voting System** is a desktop application built with **Java Swing** for the frontend and **Oracle SQL** as the backing database. It digitizes the end-to-end electoral workflow: registering voters with CNIC verification, listing candidates per election, casting ballots with strict one-vote-per-election integrity, and producing auditable tallies for administrators.

DESIGN PRINCIPLE

**One person.
One vote.
One source of truth.**

DATA INTEGRITY

Foreign keys, UNIQUE constraints, and atomic vote transactions.

SWING UI

Forms, tables and dashboards built with Java Swing components.

ORACLE BACKEND

Normalized 6-entity schema with CNIC and admin authentication.

6	1:1	4	100%
ENTITIES	VOTE RULE	MODULES	AUDITABLE

02 Core Features

01 Voter Registration
Register voters with CNIC, age, address, and assigned polling location.

02 Admin Authentication
Dedicated Admin entity with username/password to manage elections.

03 Election Management
Create elections with title, type and date; attach eligible candidates.

04 Candidate Listing
Candidates linked to elections with party affiliation and unique IDs.

05 Secure Vote Casting
Atomic vote insert with UNIQUE(VoterID, ElectionID) — no double voting.

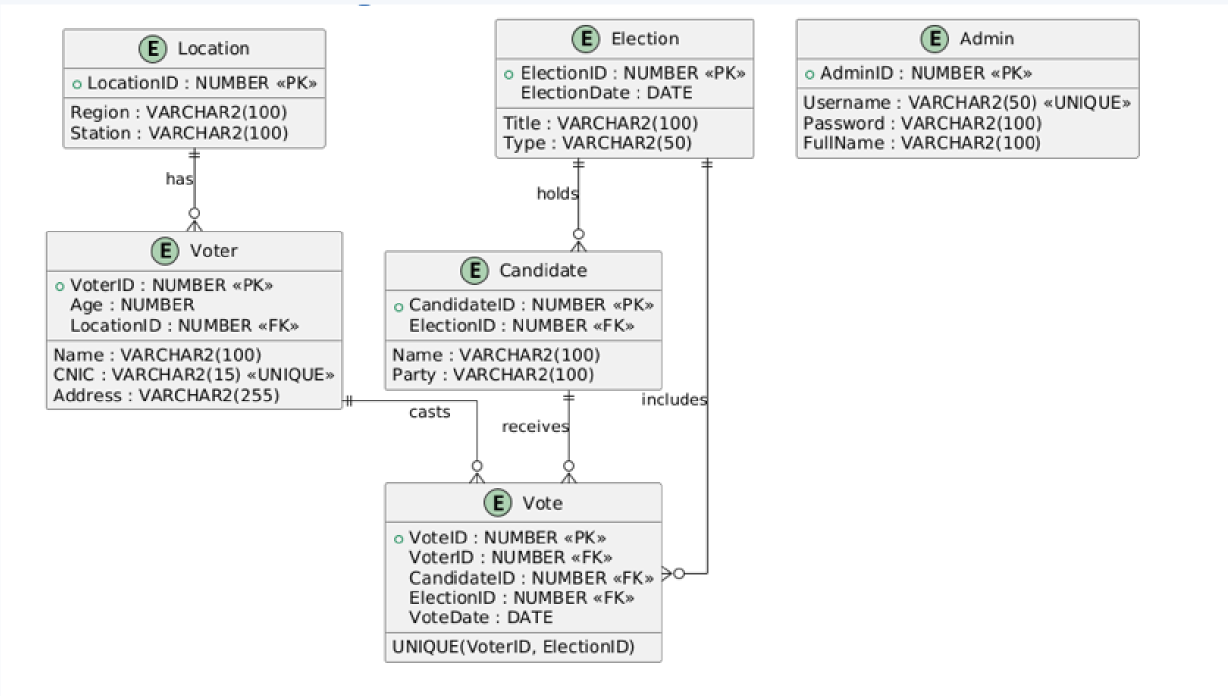
06 Polling Locations
Voters mapped to a Location (region + station) for jurisdictional control.

07 Audit-Ready Records
Every vote stamped with VoteDate; full referential integrity preserved.

08 Result Tally
Aggregate vote counts per candidate per election via SQL queries.

03 ER Diagram

Six entities — Location, Voter, Election, Candidate, Vote, Admin — modeled with strict relational constraints in Oracle SQL.



KEY CONSTRAINT

UNIQUE (VoterID, ElectionID) → one voter can cast at most one vote per election.

CNIC is UNIQUE on Voter; Admin credentials isolated; all FKs cascade-safe.

04 Stack & Architecture

TECHNOLOGY

LANGUAGE	**Java (JDK 8+)**
UI	**Java Swing**
DB DRIVER	**JDBC**
DATABASE	**Oracle SQL**
SCHEMA	**6 entities, 3NF**
AUTH	**Admin table (hashed)**
BUILD	**Manual / IDE**

Presentation Layer
Java Swing forms, tables & dialogs

↓

Application Layer
Java classes — voter / election / vote logic

↓

Data Layer
JDBC → Oracle SQL (6 normalized tables)

// vote_insert.sql

```sql
INSERT INTO Vote (VoteID, VoterID, CandidateID, ElectionID, VoteDate)
VALUES (vote_seq.NEXTVAL, :voter, :cand, :election, SYSDATE);

-- enforced by:  UNIQUE (VoterID, ElectionID)
-- ensures:      one ballot per voter per election
```